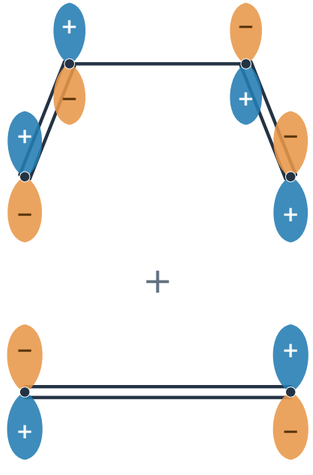
Diels–Alder [4+2] | orbital phases along the reaction

THERMALLY ALLOWED

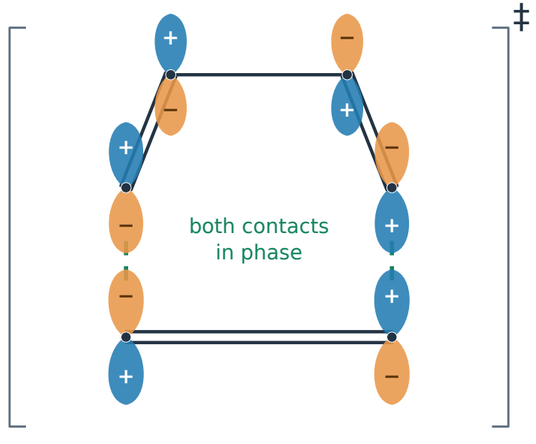
suprafacial / suprafacial

REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Diene HOMO, ψ₂ (occupied)	Hückel circuit · 6 electrons	Product connectivity



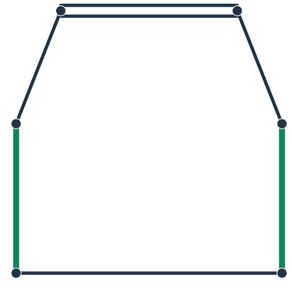
Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Δ



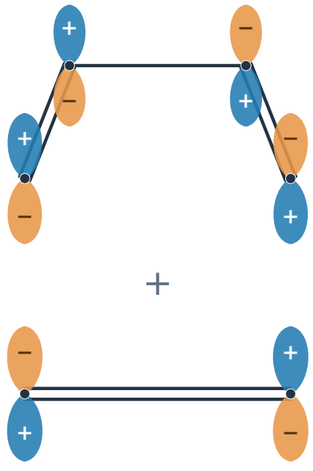
Cyclohexene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

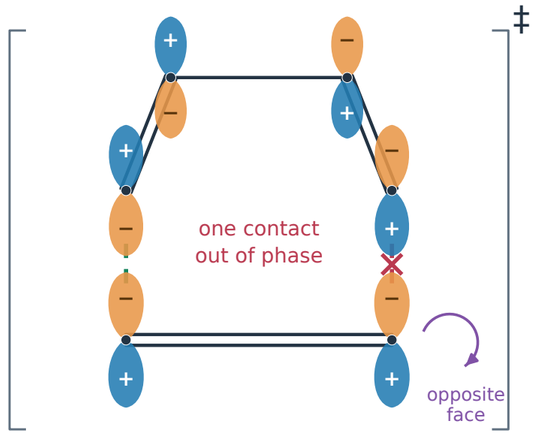
suprafacial / antarafacial

Diene HOMO, ψ₂ (occupied)	Möbius circuit · 6 electrons	Proposed product connectivity

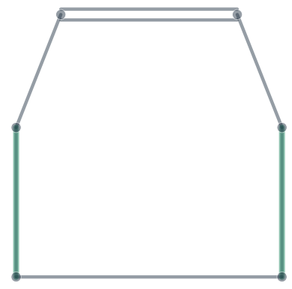


Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.



Cyclohexene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Cycloaddition panels show the diagnostic HOMO–LUMO pair. Changing one approach face changes the relative overlap sign.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

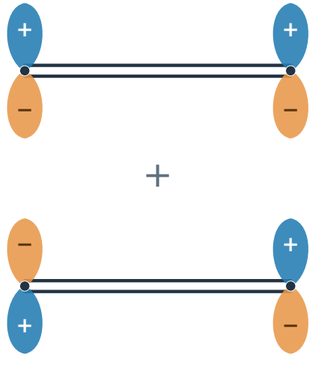
[2+2] cycloaddition | orbital phases along the reaction

THERMALLY ALLOWED (FORMAL)

suprafacial / antarafacial

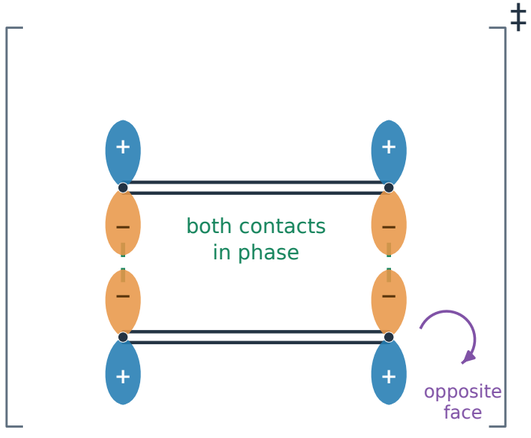
Möbius circuit · 4 electrons

Ethene HOMO, ψ₁ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Δ

Product connectivity



Cyclobutane

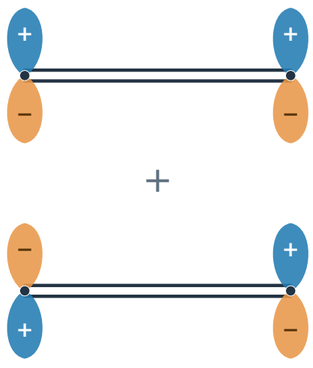
New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / suprafacial

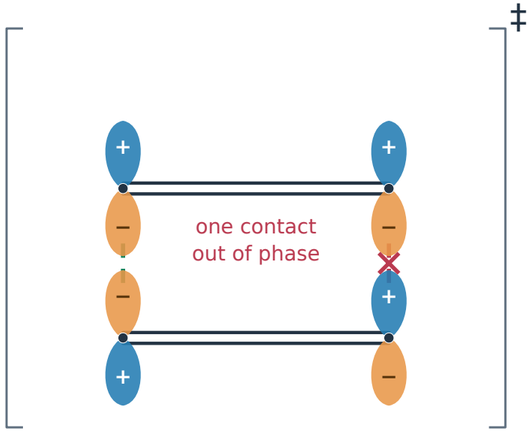
Hückel circuit · 4 electrons

Ethene HOMO, ψ₁ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Proposed product connectivity



Cyclobutane

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

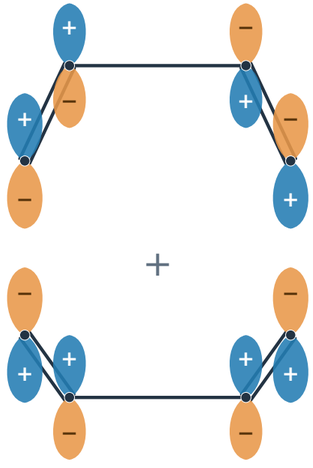
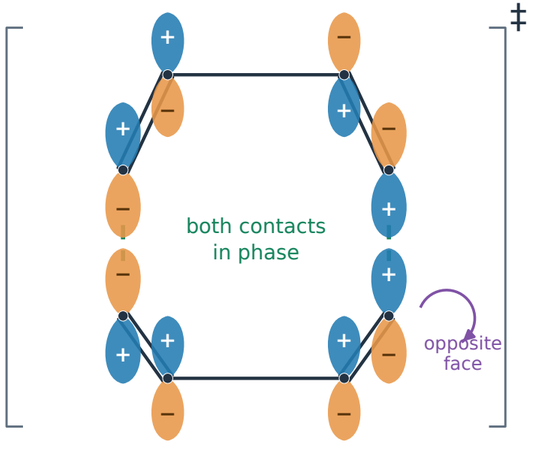
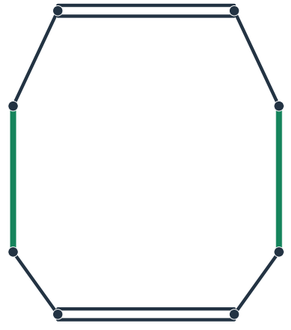
Antarafacial allowance is formal: geometry may prevent this approach, especially for ethene. Products show connectivity only.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

[4+4] cycloaddition | orbital phases along the reaction

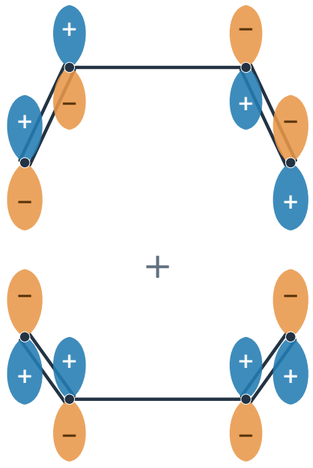
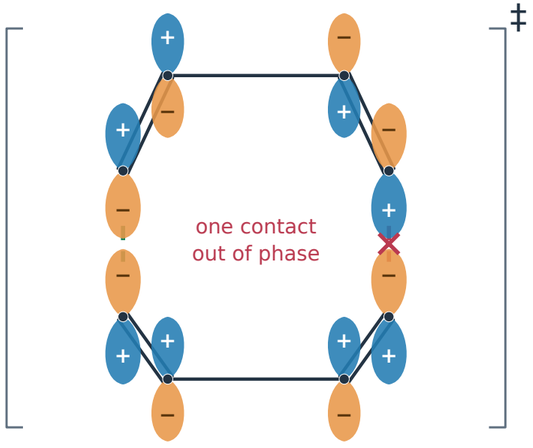
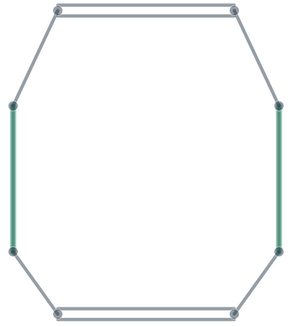
THERMALLY ALLOWED (FORMAL)

suprafacial / antarafacial

REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Diene HOMO, ψ₂ (occupied)	Möbius circuit · 8 electrons	Product connectivity
		
Diene LUMO, ψ₃ (empty)	Partial bonds join the facing lobes.	Cycloocta-1,5-diene
		New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / suprafacial

Diene HOMO, ψ₂ (occupied)	Hückel circuit · 8 electrons	Proposed product connectivity
		
Diene LUMO, ψ₃ (empty)	Partial bonds join the facing lobes.	Cycloocta-1,5-diene
		Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

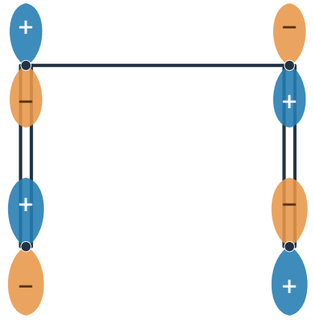
Antarafacial allowance is formal: geometry may prevent this approach, especially for ethene. Products show connectivity only.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

4π electrocyclic closure | orbital phases along the reaction

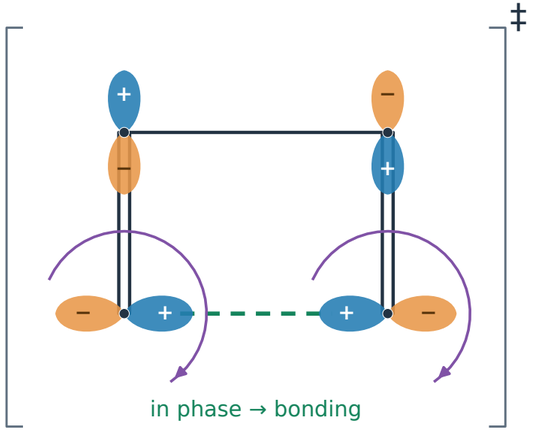
THERMALLY ALLOWED

REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Polyene HOMO, ψ₂ (occupied)	conrotatory Möbius circuit · 4 electrons	Product connectivity



Opposite terminal phases

Δ



in phase → bonding

Δ



Cyclobutene

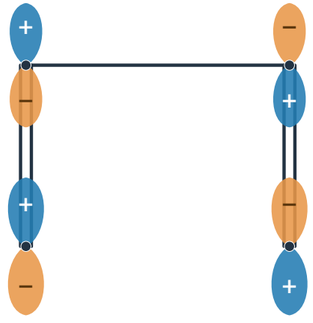
Follow the terminal lobes during rotation.

same rotation sense

New σ bond(s) highlighted in green.

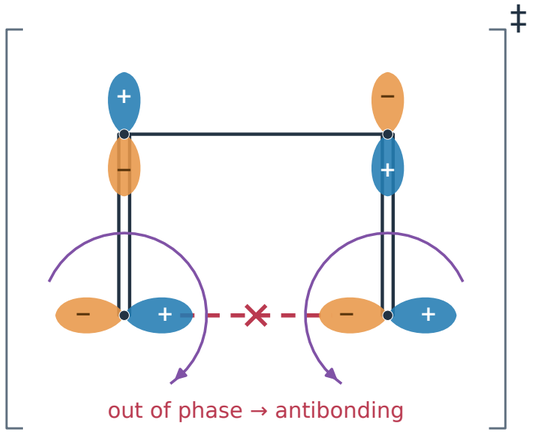
THERMALLY FORBIDDEN

Polyene HOMO, ψ₂ (occupied)	disrotatory Hückel circuit · 4 electrons	Proposed product connectivity



Opposite terminal phases

Δ



out of phase → antibonding



Cyclobutene

Follow the terminal lobes during rotation.

opposite rotation senses

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Rotation arrows follow the same viewing convention at both termini. Product connectivity is shown without substituent stereochemistry.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

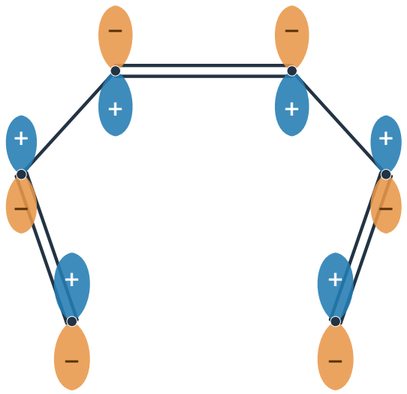
6π electrocyclic closure | orbital phases along the reaction

THERMALLY ALLOWED

disrotatory

REACTANT ORBITALS

Polyene HOMO, ψ₃ (occupied)



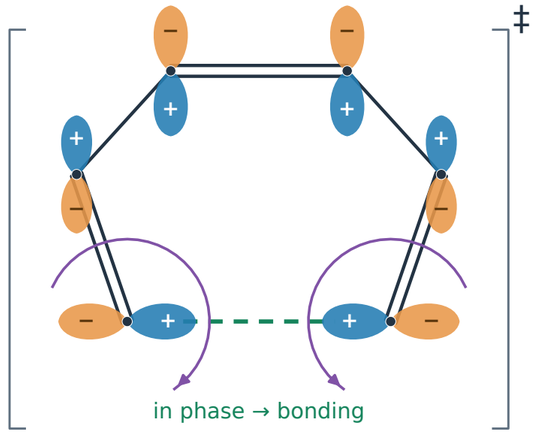
Same terminal phases

Follow the terminal lobes during rotation.

Δ

PROPOSED TRANSITION STATE

Hückel circuit · 6 electrons



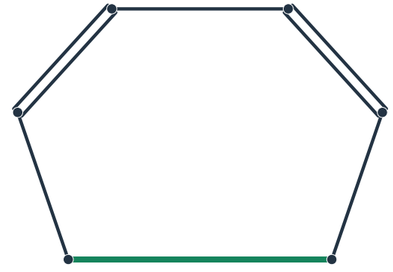
in phase → bonding

opposite rotation senses

Δ

PRODUCT

Product connectivity



Cyclohexa-1,3-diene

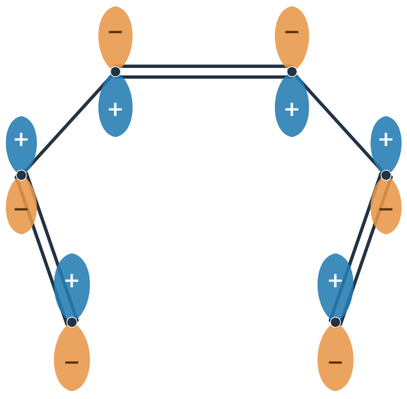
New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

conrotatory

REACTANT ORBITALS

Polyene HOMO, ψ₃ (occupied)



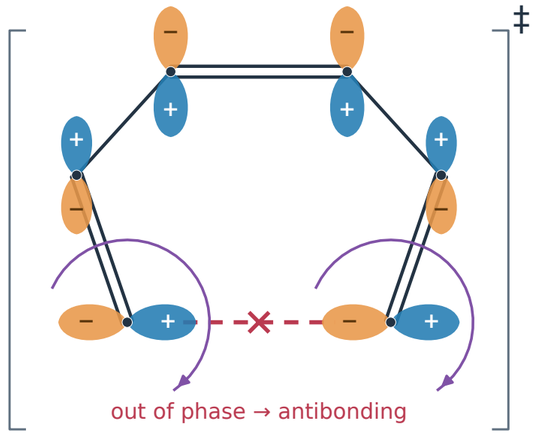
Same terminal phases

Follow the terminal lobes during rotation.

Δ

PROPOSED TRANSITION STATE

Möbius circuit · 6 electrons

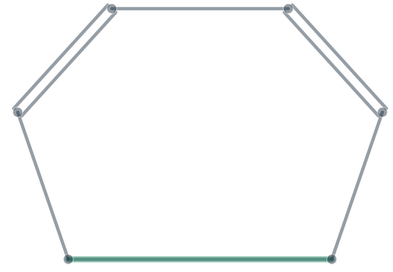


out of phase → antibonding

same rotation sense

PRODUCT

Proposed product connectivity



Cyclohexa-1,3-diene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Rotation arrows follow the same viewing convention at both termini. Product connectivity is shown without substituent stereochemistry.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

Claisen [3,3] | orbital phases along the reaction

REACTANT ORBITALS

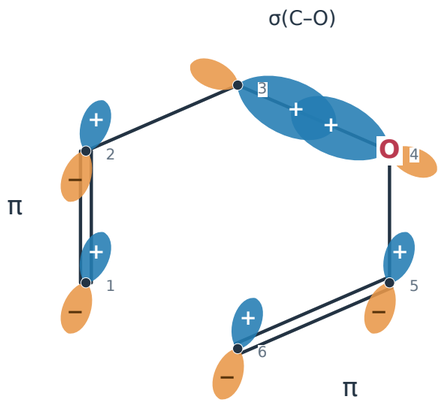
PROPOSED TRANSITION STATE

PRODUCT

THERMALLY ALLOWED

suprafacial / suprafacial

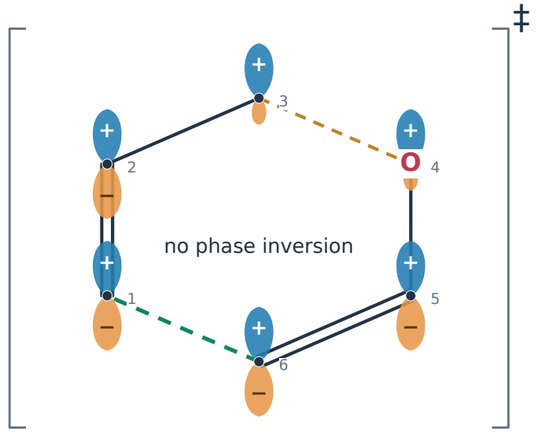
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

Δ

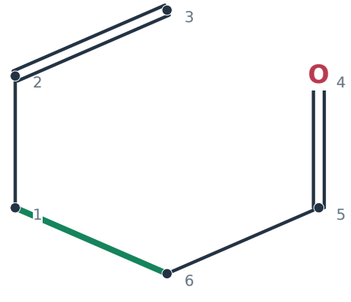
Hückel array · 6 electrons



Local overlap basis; σ centers reorganize.

Δ

Product connectivity



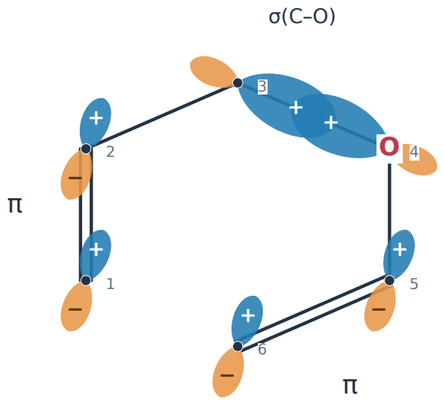
Pent-4-enal

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / antarafacial

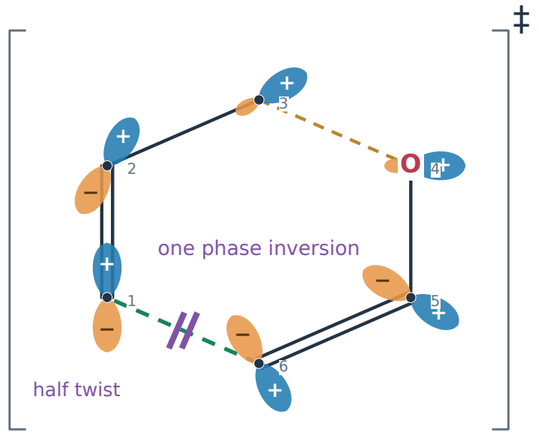
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

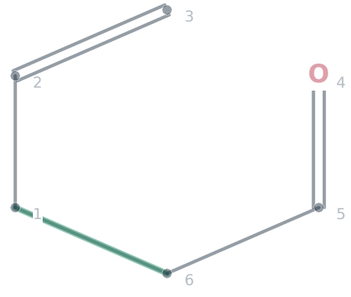
Δ

Möbius array · 6 electrons



Local overlap basis; σ centers reorganize.

Proposed product connectivity



Pent-4-enal

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

π/σ and transition-state local-basis drawings use different representations; neither is a calculated molecular HOMO.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

Cope [3,3] | orbital phases along the reaction

REACTANT ORBITALS

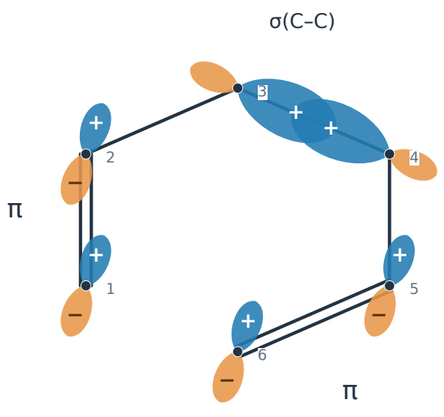
PROPOSED TRANSITION STATE

PRODUCT

THERMALLY ALLOWED

suprafacial / suprafacial

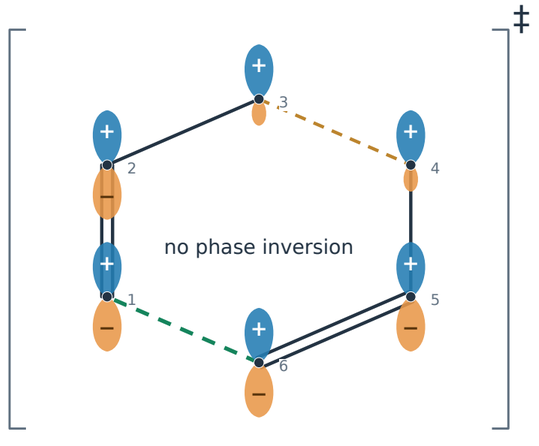
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

Δ

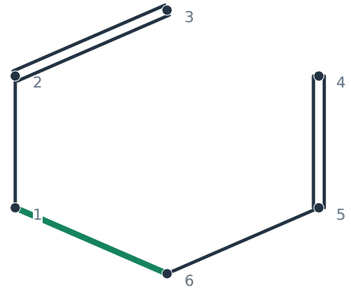
Hückel array · 6 electrons



Local overlap basis; σ centers reorganize.

Δ

Product connectivity



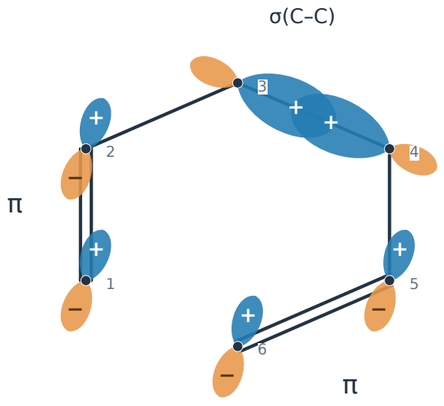
Rearranged hexa-1,5-diene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / antarafacial

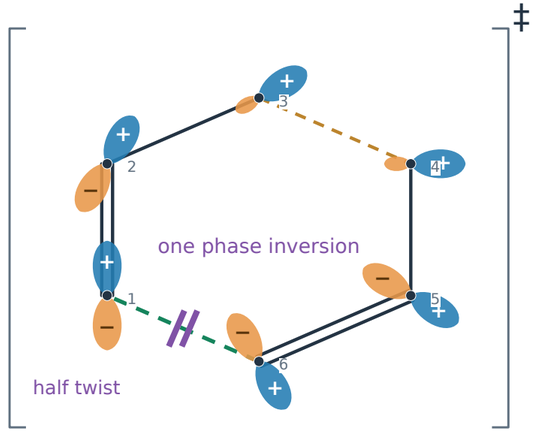
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

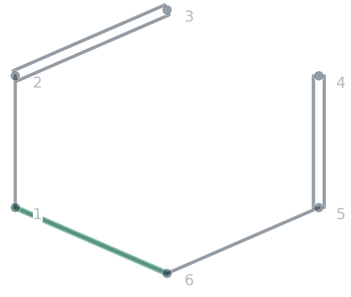
Δ

Möbius array · 6 electrons



Local overlap basis; σ centers reorganize.

Proposed product connectivity



Rearranged hexa-1,5-diene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

π/σ and transition-state local-basis drawings use different representations; neither is a calculated molecular HOMO.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.

Alder–ene | orbital phases along the reaction

REACTANT ORBITALS

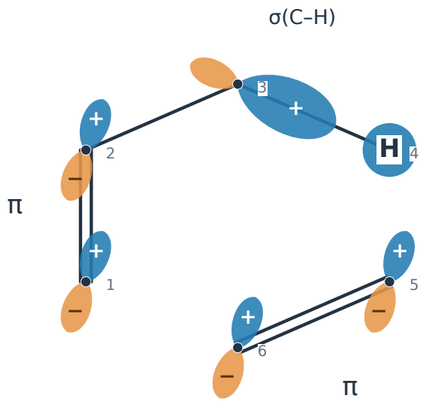
PROPOSED TRANSITION STATE

PRODUCT

THERMALLY ALLOWED

all suprafacial

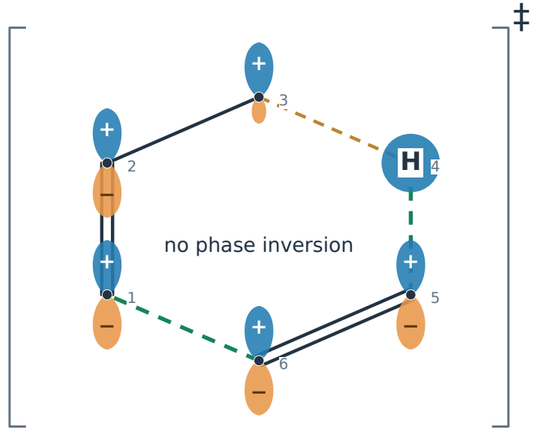
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

Δ

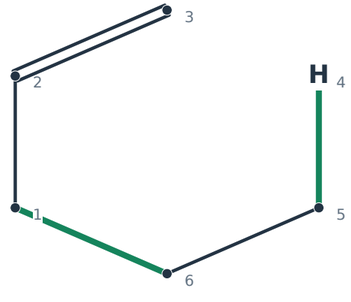
Hückel array · 6 electrons



Local overlap basis; σ centers reorganize.

Δ

Product connectivity



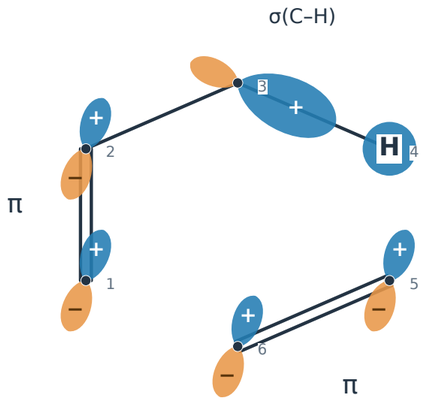
Pent-1-ene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

one antarafacial component

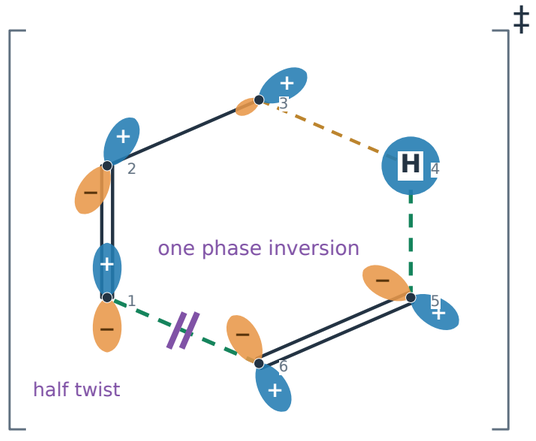
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

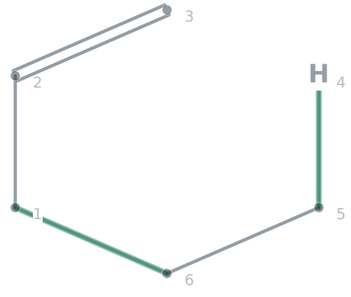
Δ

Möbius array · 6 electrons



Local overlap basis; σ centers reorganize.

Proposed product connectivity



Pent-1-ene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

π/σ and transition-state local-basis drawings use different representations; neither is a calculated molecular HOMO.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.